



DTECT PHARMACOGENETIC REPORT

COMPREHENSIVE

PATIENT INFORMATION

NAME:
ACC #:
SAMPLE ID:
DOB:
GENDER:

SPECIMEN DETAILS

SPECIMEN TYPE:
COLLECTION DATE:
RECEIVED DATE:
REPORT DATE:

FACILITY

ORDERING PHYSICIAN:
FACILITY

NAME:
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1 INTERPRETATION GUIDELINES

1.1 Purpose of Pharmacogenetic Report

This pharmacogenetic (PGx) report explains how a person's inherited genetic variation may influence their response to commonly prescribed medicines. Genetic differences can affect how drugs are absorbed, metabolised, distributed, or eliminated, and may therefore influence treatment effectiveness, the risk of side effects, or the most appropriate dose. The information presented here supports, but does not replace, clinical judgement.

This report has been developed in accordance with internationally recognised pharmacogenetic standards and recommendations, primarily from the Clinical Pharmacogenetics Implementation Consortium (CPIC), ClinPGx, the Pharmacogenetics Research Network (PGRN), PharmVar, and professional pharmacy and clinical pharmacology organisations.

1.2 GUIDANCE LEVELS

The guidance levels provide an immediate, clinically relevant summary of how strongly a patient's genetic results may affect the prescribing of a particular medicine.

DRUGS WITH NORMAL PRESCRIPTIONS		DRUGS THAT HAVE ADDITIONAL INFORMATION	
	These medicines are not expected to be affected by the patient's genetic test results.		For these medicines, a genetic variant has been detected that is known to cause a <i>mild</i> change in drug response.
	Standard prescribing and routine clinical monitoring are appropriate. No genetically based dose adjustment or special precautions are required.		This may include a slightly altered metabolic rate or a small increase in the likelihood of minor side effects. In most cases, the drug remains safe and effective when used at standard doses.
DRUGS TO BE USED WITH CAUTION		DRUGS TO BE REPLACED OR MONITORED	
	These medicines are associated with genetic variants that <i>moderately</i> affect drug metabolism or response.		The information is provided to support awareness and shared decision-making between the clinician and the patient.
	As a consequence, some patients may require a dose adjustment, closer monitoring for side effects, or specific clinical precautions.		For these medicines, the identified genetic variants are known to <i>substantially</i> alter drug response.
The drug may still be prescribed, but enhanced clinical supervision is recommended.		This may result in unpredictable drug levels, reduced benefit, or an increased risk of serious adverse reactions.	
		Where possible, consider an alternative medication not affected by the same genetic pathway.	
		If no suitable alternative exists, intensive monitoring (such as therapeutic drug monitoring) may be required.	

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1.3 EVIDENCE STRENGTH

Level	Evidence	Description
1A	STRONG	Derived from well-designed and replicated studies showing a clear and consistent association between genetic variation and clinical outcome. Recommendations at this level are considered highly reliable for clinical implementation, particularly when endorsed by organisations such as CPIC or the Dutch Pharmacogenetics Working Group (DPWG).
1B		
2A	MODERATE	Supported by clinical data demonstrating an association with drug response, but with some limitations in study size, consistency, or generalisability. These findings are clinically useful but should be interpreted alongside patient-specific factors.
2B		
3	WEAK	Based on limited or inconsistent data. These findings are provided for informational purposes only and should not be the sole basis for clinical decision-making.

ACTIONABLE

Recommendations are based upon publications by international pharmacogenetic expert groups, consortia, or regulatory bodies (CPIC, DPWG, FDA, EMA). Recommendations are suitable for implementation in a clinical setting. Guidelines may change as new knowledge arises.

INFORMATIVE

There are insufficient or contradictory findings documenting the impact of a given genetic polymorphism or drug interaction. Recommendations are informative, and implementation in a clinical setting is optional.

1.4 PRIMARY SOURCES OF KNOWLEDGE

Primary sources provide the evidence supporting the drug-gene association and the recommended actions. The selection of the primary source depends on the recommendation's specificity and the source's rank. The FDA ranks highest, followed by clinical practice guidelines and the DPWG's internal review of published literature.

Acronym	Full Name	Primary Role in Pharmacogenetics
ACMG	American College of Medical Genetics and Genomics	Develops technical standards for laboratory testing and variant classification.
ASCPT	American Society for Clinical Pharmacology and Therapeutics	Endorses clinical guidelines and publishes peer-reviewed research in therapeutics.
ASHP	American Society of Health-System Pharmacists	Provides clinical guidelines and education for pharmacists in genomic medicine.
ClinPGx	Clinical Pharmacogenomics Resource	Integrates PharmGKB, CPIC, and PharmCAT to provide curated genomic evidence.
CPIC	Clinical Pharmacogenetics Implementation Consortium	Creates evidence-based, actionable guidelines for drug-gene pairs.
DPWG	Dutch Pharmacogenetics Working Group	Develops European-based pharmacogenetic guidelines and dosing recommendations.
EMA	European Medicines Agency	European regulatory body that includes genomic information in drug labels.
Eu-PIC	European Pharmacogenetics Implementation Consortium	Drives the clinical implementation of pharmacogenetics across Europe.
FDA	U.S. Food and Drug Administration	A regulatory body that provides pharmacogenetic associations and safety labelling.
PharmVar	Pharmacogene Variation Consortium	The central authority for naming and categorising genetic variants (alleles).
PGRN	Pharmacogenomics Research Network	Catalyses global research in precision medicine and genomic drug response.

2 CURRENT MEDICATION ANALYSIS

The Current Medication Analysis section provides a cross-reference of the medications the patient is currently taking in relation to their unique genetic profile. This analysis flags potential gene-drug interactions that may already be affecting the patient's therapeutic outcomes by diminishing efficacy, causing unrecognised side effects, or altering drug exposure in a clinically significant way.

2.1 How to Interpret the Analysis Table

The table presented in this section is designed for clarity and rapid comprehension. Each row corresponds to one of the patient's current medications, and the columns provide layered information to guide clinical assessment.

- **Drug:** This column lists the generic name of the medication as provided on the test requisition form.
- **Gene:** This column identifies the specific gene for which the patient's genetic makeup may alter their response to the indicated drug. If no clinically relevant gene-drug interaction is known for a medication, or if the patient's genetics for relevant genes are normal, this column may indicate 'N/A'.
- **Effect:** This provides a high-level summary of the patient's predicted metabolic status or response related to the drug, such as "Normal Metaboliser" or "Slightly Slower Metaboliser." This directly reflects the diplotype identified in Section 5 of this report.
- **Advice/Effect Information:** This column delivers a concise clinical summary of the potential impact. It offers initial guidance, explains the consequences of the genetic finding (e.g., "may have reduced clearance rates"), and suggests a course of action, such as monitoring for toxicity or considering dose adjustments.

2.2 The Colour-Coding System

To further enhance rapid interpretation, this section uses a colour-coded system that provides an immediate visual cue for the level of clinical concern. This system directly corresponds to the detailed guidance levels defined on page 2 of the report. Understanding this colour-to-action mapping is essential to use the results efficiently.

2.3 Results

Current Medicines have not been provided.

3 SUMMARY RESULTS

The Summary Results section focuses on proactive, prospective therapeutic planning. This extensive section functions as a personalised formulary, meticulously crafted based on the patient's unique genetic profile. Its purpose is to help clinicians anticipate how patients are likely to respond to a wide array of medications before writing a prescription.

By consulting this summary, clinicians can move beyond the conventional trial-and-error paradigm of medication selection. Instead, they can utilise genetic information to take a data-driven approach, selecting therapies most likely to be effective and safe for that individual, while avoiding those that pose a genetically increased risk of failure or harm. This proactive strategy is the cornerstone of personalised medicine, aiming to improve outcomes, reduce adverse drug events, and improve care efficiency.

DRUG MAIN GROUP	DRUG SUB-GROUP	DRUGS WITH NORMAL PRESCRIPTIONS	DRUGS THAT HAVE ADDITIONAL INFORMATION	DRUGS TO BE USED WITH CAUTION	DRUGS TO BE REPLACED OR MONITORED
					
ALIMENTARY TRACT AND METABOLISM	STOMATOLOGICAL PREPARATIONS	Dexamethasone Metronidazole Minocycline			
ALIMENTARY TRACT AND METABOLISM	DRUGS FOR PEPTIC ULCER AND GASTRO-OESOPHAGEAL REFLUX DISEASE (GORD)		Esomeprazole Rabeprazole	Dexlansoprazole Lansoprazole Omeprazole Pantoprazole	
ALIMENTARY TRACT AND METABOLISM	PROPULSIVES	Itopride			
ALIMENTARY TRACT AND METABOLISM	ANTIEMETICS AND ANTINAUSEANTS	Dolasetron Granisetron Palonosetron Tropisetron	Ondansetron		
ALIMENTARY TRACT AND METABOLISM	DRUGS FOR CONSTIPATION		Naloxone		
ALIMENTARY TRACT AND METABOLISM	INTESTINAL ANTIINFLAMMATORY AGENTS	Budesonide Prednisolone Prednisone	Sulfasalazine		
ALIMENTARY TRACT AND METABOLISM	ANTIOBESITY PREPARATIONS		Sibutramine		
ALIMENTARY TRACT AND METABOLISM	BLOOD GLUCOSE LOWERING DRUGS	Sitagliptin Tolbutamide Troglitazone Vildagliptin	Gliclazide Glimepiride Glipizide Liraglutide Metformin Pioglitazone Repaglinide Rosiglitazone		
ALIMENTARY TRACT AND METABOLISM	OTHER ALIMENTARY TRACT AND METABOLISM PRODUCTS	Cysteamine Migalastat Zinc Acetate			
BLOOD AND BLOOD FORMING ORGANS	ANTITHROMBOTIC AGENTS	Cangrelor Cilostazol Coumarin Dabigatran Heparin Ticlopidine	Apixaban Fluindione Prasugrel Rivaroxaban Ticagrelor	Phenprocoumon	Acenocoumarol Clopidogrel Warfarin
BLOOD AND BLOOD FORMING ORGANS	VITAMIN B12 AND FOLIC ACID	Cyanocobalamin Folic Acid			
CARDIOVASCULAR SYSTEM	CARDIAC GLYCOSIDES		Digoxin		
CARDIOVASCULAR SYSTEM	ANTIARRHYTHMICS	Amiodarone Disopyramide Flecainide Mexiletine			

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		Propafenone Quinidine			
CARDIOVASCULAR SYSTEM	CARDIAC STIMULANTS EXCL. CARDIAC GLYCOSIDES	Ephedrine Phenylephrine	Dobutamine		
CARDIOVASCULAR SYSTEM	OTHER CARDIAC PREPARATIONS	Ibuprofen Regadenoson			
CARDIOVASCULAR SYSTEM	ANTIADRENERGIC AGENTS	Debrisoquine	Clonidine		
CARDIOVASCULAR SYSTEM	ARTERIOLAR SMOOTH MUSCLE		Hydralazine		
CARDIOVASCULAR SYSTEM	LOW-CEILING DIURETICS	Chlorthalidone	Hydrochlorothiazide		
CARDIOVASCULAR SYSTEM	HIGH-CEILING DIURETICS	Bumetanide Furosemide Torsemide			
CARDIOVASCULAR SYSTEM	POTASSIUM-SPARING AGENTS		Spironolactone		
CARDIOVASCULAR SYSTEM	AGENTS FOR TREATMENT OF HEMORRHOIDS AND ANAL FISSURES FOR TOPICAL USE	Dexamethasone Prednisolone Triamcinolone			
CARDIOVASCULAR SYSTEM	BETA BLOCKING AGENTS	Bisoprolol Propranolol Talinolol Timolol	Atenolol Carvedilol Metoprolol		
CARDIOVASCULAR SYSTEM	SELECTIVE CALCIUM CHANNEL BLOCKERS WITH MAINLY VASCULAR EFFECTS	Amlodipine Nimodipine Nitrendipine	Nifedipine		
CARDIOVASCULAR SYSTEM	SELECTIVE CALCIUM CHANNEL BLOCKERS WITH DIRECT CARDIAC EFFECTS		Verapamil		
CARDIOVASCULAR SYSTEM	ACE INHIBITORS	Captopril Imidapril Lisinopril Trandolapril	Benazepril Enalapril Perindopril Quinapril Ramipril		
CARDIOVASCULAR SYSTEM	ANGIOTENSIN II RECEPTOR BLOCKERS (ARBs)		Candesartan Irbesartan Losartan Telmisartan		
CARDIOVASCULAR SYSTEM	LIPID MODIFYING AGENTS	Cerivastatin Fluvastatin Pitavastatin	Atorvastatin Fenofibrate Lovastatin Pravastatin Simvastatin	Rosuvastatin	
DERMATOLOGICAL DRUGS	ANTIPRURITICS			Doxepin	
DERMATOLOGICAL DRUGS	ANTIPSORIATICS FOR SYSTEMIC USE	Methoxsalen			
DERMATOLOGICAL DRUGS	ANTIBIOTICS FOR TOPICAL USE	Gentamicin			
DERMATOLOGICAL DRUGS	CHEMOTHERAPEUTICS FOR TOPICAL USE	Imiquimod Metronidazole			
DERMATOLOGICAL DRUGS	CORTICOSTEROIDS	Budesonide Dexamethasone Methylprednisolone Prednisolone Triamcinolone			
DERMATOLOGICAL DRUGS	CORTICOSTEROIDS	Dexamethasone Prednisolone			

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		Triamcinolone			
DERMATOLOGICAL DRUGS	ANTISEPTICS AND DISINFECTANTS		Ethanol		
DERMATOLOGICAL DRUGS	ANTI-ACNE PREPARATIONS FOR TOPICAL USE	Clindamycin Dapsone Dexamethasone Erythromycin Methylprednisolone Minocycline			
DERMATOLOGICAL DRUGS	OTHER DERMATOLOGICAL PREPARATIONS	Diclofenac	Caffeine		Tacrolimus
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	ANTIINFECTIVES AND ANTISEPTICS	Clindamycin Metronidazole			
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	OTHER GYNECOLOGICALS	Ibuprofen Ritodrine			
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	ESTROGENS	Conjugated Estrogens Estradiol			
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	PROGESTOGENS	Medroxyprogesterone			
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	GONADOTROPINS AND OTHER OVULATION STIMULANTS	Follitropin Beta Urofollitropin			
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	OTHER SEX HORMONES AND MODULATORS OF THE GENITAL SYSTEM	Raloxifene			
GENITOURINARY SYSTEM AND REPRODUCTIVE HORMONES	UROLOGICALS	Dapoxetine Fesoterodine Tolterodine Vardenafil	Sildenafil		
SYSTEMIC HORMONAL PREPARATIONS	ANTERIOR PITUITARY LOBE HORMONES AND ANALOGUES	Thyrotropin Alfa			
SYSTEMIC HORMONAL PREPARATIONS	CORTICOSTEROIDS FOR SYSTEMIC USE	Dexamethasone Methylprednisolone Prednisolone Prednisone Triamcinolone			
SYSTEMIC HORMONAL PREPARATIONS	ANTITHYROID PREPARATIONS	Carbimazole	Propylthiouracil		
ANTIINFECTIVES FOR SYSTEMIC USE	TETRACYCLINES	Minocycline			
ANTIINFECTIVES FOR SYSTEMIC USE	BETA-LACTAM ANTIBACTERIALS	Amoxicillin Dicloxacillin Flucloxacillin Penicillin G			
ANTIINFECTIVES FOR SYSTEMIC USE	OTHER BETA-LACTAM ANTIBACTERIALS	Cefotaxime Ceftriaxone			
ANTIINFECTIVES FOR SYSTEMIC USE	SULFONAMIDES AND TRIMETHOPRIM		Sulfamethoxazole		
ANTIINFECTIVES FOR SYSTEMIC USE	MACROLIDES	Clindamycin Erythromycin			
ANTIINFECTIVES FOR SYSTEMIC USE	AMINOGLYCOSIDE ANTIBACTERIALS	Amikacin Ciprofloxacin Gentamicin Isepamicin Kanamycin			

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		Neomycin Streptomycin			
ANTIINFECTIVES FOR SYSTEMIC USE	OTHER ANTIBACTERIALS	Chloramphenicol Metronidazole Nitrofurantoin	Daptomycin		
ANTIINFECTIVES FOR SYSTEMIC USE	ANTIMYCOTICS FOR SYSTEMIC USE			Voriconazole	
ANTIINFECTIVES FOR SYSTEMIC USE	DRUGS FOR TREATMENT OF TUBERCULOSIS	Ethambutol Pyrazinamide	Isoniazid		
ANTIINFECTIVES FOR SYSTEMIC USE	DIRECT ACTING ANTIVIRALS	Abacavir Amprenavir Boceprevir Daclatasvir Dolutegravir Etravirine Faldaprevir Lamivudine Letermovir Maraviroc Simeprevir Sofosbuvir Valganciclovir Zidovudine	Atazanavir Didanosine Indinavir Nelfinavir Nevirapine Oseltamivir Ribavirin Ritonavir Stavudine Telaprevir		Efavirenz
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	ALKYLATING AGENTS	Chlorambucil Temozolomide Thiotepa	Busulfan Cyclophosphamide Melphalan		
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	ANTIMETABOLITES	Cladribine Fludarabine Raltitrexed Tegafur Thioguanine	Cytarabine Gemcitabine Mercaptopurine Methotrexate Pemetrexed	Capecitabine Fluorouracil	
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	PLANT ALKALOIDS AND OTHER NATURAL PRODUCTS	Etoposide Topotecan Vindesine Vinorelbine	Docetaxel Irinotecan Paclitaxel Vincristine		
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	CYTOTOXIC ANTIBIOTICS AND RELATED SUBSTANCES	Bleomycin Epirubicin Mitoxantrone	Daunorubicin Doxorubicin Idarubicin		
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	PROTEIN KINASE INHIBITORS	Axitinib Dasatinib Erlotinib Lapatinib Nilotinib Pazopanib Temsilolimus	Everolimus Gefitinib Imatinib Sorafenib Sunitinib		
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	OTHER ANTINEOPLASTIC AGENTS	Asparaginase Atrasentan Bortezomib Carboplatin Gemtuzumab Ozogamicin Panitumumab Pegaspargase Rituximab Trastuzumab	Bevacizumab Celecoxib Cetuximab Cisplatin Mitotane Oxaliplatin		
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	HORMONES AND RELATED AGENTS	Medroxyprogesterone			
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	HORMONE ANTAGONISTS AND RELATED AGENTS		Anastrozole Exemestane Letrozole Tamoxifen		

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ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	IMMUNOSTIMULANTS	Glatiramer Acetate Interferon Alfa-2A Interferon Alfa-2B Interferon Beta-1B Peginterferon Alfa-2A	Interferon Beta-1A		
ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	IMMUNOSUPPRESSANTS	Certolizumab Pegol Cladribine Dimethyl Fumarate Eculizumab Infliximab Lenalidomide Siponimod Thalidomide	Adalimumab Azathioprine Cyclosporine Etanercept Everolimus Leflunomide Methotrexate Mycophenolic Acid Sirolimus Tocilizumab Ustekinumab		Tacrolimus
MUSCULOSKELETAL SYSTEM	ANTIINFLAMMATORY AND ANTIRHEUMATIC PRODUCTS	Diclofenac Flurbiprofen Ibuprofen Ketoprofen Ketorolac Lornoxicam Meloxicam Naproxen Piroxicam Sulindac Tenoxicam	Celecoxib		
MUSCULOSKELETAL SYSTEM	TOPICAL PRODUCTS FOR JOINT AND MUSCULAR PAIN	Diclofenac Flurbiprofen Ibuprofen Ketoprofen Piroxicam			
MUSCULOSKELETAL SYSTEM	ANTIGOUT PREPARATIONS	Febuxostat Pegloticase Sulfinpyrazone	Allopurinol		
MUSCULOSKELETAL SYSTEM	DRUGS AFFECTING BONE STRUCTURE AND MINERALIZATION	Alendronate Etidronic Acid Risedronate Tiludronate			
MUSCULOSKELETAL SYSTEM	OTHER DRUGS FOR DISORDERS OF THE MUSCULO-SKELETAL SYSTEM	Ataluren	Succinylcholine		
NERVOUS SYSTEM	ANESTHETICS	Desflurane Enflurane Halothane Isoflurane Ketamine Nitrous Oxide Propofol Remifentanyl Trichloroethylene	Alfentanil Fentanyl Sevoflurane Sufentanil		
NERVOUS SYSTEM	OPIOIDS	Butorphanol Dihydrocodeine Tapentadol	Buprenorphine Morphine Oxycodone Tramadol		
NERVOUS SYSTEM	OTHER ANALGESICS AND ANTIPYRETICS	Dipyrrone Methoxyflurane Salicylamide	Acetaminophen Aspirin		
NERVOUS SYSTEM	ANTIMIGRAINE PREPARATIONS		Clonidine Sumatriptan		
NERVOUS SYSTEM	ANTIEPILEPTICS	Brivaracetam	Carbamazepine		

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		✓	💡	!	✗
		Levetiracetam Mephenytoin Zonisamide	Gabapentin Lamotrigine Oxcarbazepine Phenobarbital Phenytoin Pregabalin Topiramate Valproic Acid		
NERVOUS SYSTEM	DOPAMINERGIC AGENTS	Amantadine Entacapone	Levodopa Pramipexole Rasagiline		
NERVOUS SYSTEM	ANTIPSYCHOTICS	Amisulpride Chlorpromazine Fluphenazine Haloperidol Iloperidone Lurasidone Paliperidone Perphenazine Prochlorperazine Thioridazine Trifluoperazine Ziprasidone Zuclopenthixol	Aripiprazole Clozapine Lithium Olanzapine Quetiapine Risperidone		
NERVOUS SYSTEM	ANXIOLYTICS	Alprazolam Diazepam Lorazepam	Clobazam		
NERVOUS SYSTEM	HYPNOTICS AND SEDATIVES		Dexmedetomidine Midazolam		
NERVOUS SYSTEM	ANTIDEPRESSANTS	Agomelatine Desipramine Fluvoxamine Mianserin Mirtazapine Nefazodone Vortioxetine	Bupropion Duloxetine Fluoxetine L-Tryptophan Milnacipran Nortriptyline Paroxetine Tianeptine Venlafaxine	Amitriptyline Citalopram Clomipramine Doxepin Escitalopram Imipramine Sertraline Trimipramine	
NERVOUS SYSTEM	PSYCHOSTIMULANTS	Amphetamine Dextroamphetamine Modafinil	Atomoxetine Caffeine Methylphenidate		
NERVOUS SYSTEM	ANTI-DEMENTIA DRUGS	Donepezil Galantamine	Memantine Rivastigmine		
NERVOUS SYSTEM	DRUGS USED IN ADDICTIVE DISORDERS	Acamprosate Varenicline	Buprenorphine Disulfiram Methadone Naltrexone Nicotine		
ANTIPARASITIC INCLUDING INSECTICIDES AND REPELLENTS	AGENTS AGAINST AMOEBIASIS AND OTHER PROTOZOAL DISEASES	Metronidazole			
ANTIPARASITIC INCLUDING INSECTICIDES AND REPELLENTS	ANTIMALARIALS	Amodiaquine Artemether Artesunate Hydroxychloroquine Primaquine Pyrimethamine			
ANTIPARASITIC INCLUDING INSECTICIDES AND REPELLENTS	ECTOPARASITICIDES		Disulfiram		
RESPIRATORY SYSTEM	DECONGESTANTS AND OTHER NASAL	Budesonide Dexamethasone	Flunisolide		

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		✓	💡	!	✗
	PREPARATIONS FOR TOPICAL USE	Ephedrine Nedocromil Phenylephrine Prednisolone Triamcinolone			
RESPIRATORY SYSTEM	ADRENERGICS	Salbutamol Salmeterol Terbutaline			
RESPIRATORY SYSTEM	OTHER DRUGS FOR OBSTRUCTIVE AIRWAY DISEASES	Budesonide Nedocromil Triamcinolone	Flunisolide		
RESPIRATORY SYSTEM	ADRENERGICS FOR SYSTEMIC USE	Ephedrine Salbutamol Terbutaline			
RESPIRATORY SYSTEM	OTHER SYSTEMIC DRUGS FOR OBSTRUCTIVE AIRWAY DISEASES	Zafirlukast	Montelukast		
RESPIRATORY SYSTEM	EXPECTORANTS	Acetylcysteine Carbocisteine			
RESPIRATORY SYSTEM	COUGH SUPPRESSANTS	Dextromethorphan Ethylmorphine	Codeine Hydrocodone		
RESPIRATORY SYSTEM	ANTIHISTAMINES FOR SYSTEMIC USE	Fexofenadine Mizolastine	Desloratadine		
RESPIRATORY SYSTEM	OTHER RESPIRATORY SYSTEM PRODUCTS	Ivacaftor Lumacaftor Tezacaftor			
SENSORY ORGANS	ANTIINFECTIVES	Dihydrostreptomycin Erythromycin Gentamicin Micronomicin Trifluridine			
SENSORY ORGANS	ANTIINFLAMMATORY AGENTS	Dexamethasone Diclofenac Flurbiprofen Ketorolac Piroxicam Prednisolone Triamcinolone			
SENSORY ORGANS	ANTIINFLAMMATORY AGENTS AND ANTIINFECTIVES IN COMBINATION	Dexamethasone Prednisolone			
SENSORY ORGANS	ANTI GLAUCOMA PREPARATIONS AND MIOTICS	Acetazolamide Timolol	Clonidine Latanoprost Methazolamide		
SENSORY ORGANS	MYDRIATICS AND CYCLOPLEGICS	Ephedrine Phenylephrine			
SENSORY ORGANS	DECONGESTANTS AND ANTIALLERGICS	Nedocromil Phenylephrine			
SENSORY ORGANS	OCULAR VASCULAR DISORDER AGENTS	Pegaptanib Ranibizumab			
SENSORY ORGANS	OTHER OPHTHALMOLOGICALS	Acetylcysteine Ethylmorphine	Sirolimus		
SENSORY ORGANS	ANTIINFECTIVES	Gentamicin			
SENSORY ORGANS	CORTICOSTEROIDS	Dexamethasone Prednisolone			
SENSORY ORGANS	ANTIINFECTIVES	Gentamicin			
SENSORY ORGANS	CORTICOSTEROIDS	Dexamethasone Prednisolone			
VARIOUS ANATOMICAL STRUCTURES	OTHER DIAGNOSTIC AGENTS	Dexrazoxane Dimercaprol	Deferasirox Deferiprone		

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		✓	💡	!	✗
		Folic Acid Methacholine Tolbutamide			

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4 DOCTOR'S SUMMARY INFORMATION

The Doctor's Summary Information is a critical intermediate section that bridges the high-level summary with the deep-dive evidence. This summary provides a more granular, clinician-focused explanation of the nature of concern. Its purpose is to answer the crucial follow-up question: "If this medication is a potential issue, is it a matter of safety, efficacy, or dosing?"

This section translates complex genetic findings into clinically relevant categories, helping physicians anticipate the specific problem that may arise. This focused information is essential for tailoring a patient's monitoring plan and management strategy.

4.1 Interpreting the Summary Table and the "Effect" Column

The table in this section organises drugs with moderate-to-high-level genetic interactions. It includes standard reference information such as the Generic Name, Trade Name(s), ATC code, and Clinical Group for easy identification. The most clinically significant column is "Effect", as it specifies the primary way in which the patient's genetics are predicted to influence their response to the drug.

"Effect" Term	Clinical Questions: It Answers	Potential Clinical Action and Monitoring Strategy
Toxicity	Is there an increased risk of adverse drug reactions (ADRs), side effects, or harm due to impaired clearance or other genetic factors?	Proactively monitor for specific, known ADRs associated with the drug. Consider a lower starting dose. Educate the patient on key symptoms to report immediately.
Efficacy	Is there an increased risk that the drug will not produce the desired therapeutic effect at standard doses?	Monitor closely for therapeutic failure or a suboptimal response. Consider a higher dose if deemed safe, or select an alternative drug if clinical goals are not met. Manage patient expectations regarding treatment outcomes.
Dosage	Is the standard, approved dosage for this drug likely to be inappropriate for this patient, requiring a modification?	Consult detailed guidelines from primary sources (e.g., CPIC, DPWG) for specific dose adjustment recommendations (either an increase or a decrease). Consider therapeutic drug monitoring (TDM) to guide dosing.
Metabolism/PK	How does this patient's body absorb, distribute, metabolise, or excrete the drug differently from the general population?	This term describes the underlying pharmacokinetic mechanism that often leads to the observed effects on toxicity, efficacy, or dosing. It provides the physiological rationale for the clinical recommendation.

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4.2 Doctor's Summary Results

Name	Trade Name	ATC	ATC Group	ATC Subgroup	Gene	Impact	Evidence	Clinical Effect
Acenocoumarol	Acenocoumarol	B01AA07	BLOOD AND BLOOD FORMING ORGANS	Vitamin K antagonists	VKORC1	Dosage Toxicity	1A	May require decreased acenocoumarol dose. May have increased likelihood of Hemorrhage.
Clopidogrel	Duoplavin Plavix Zyllt	B01AC04	BLOOD AND BLOOD FORMING ORGANS	Platelet aggregation inhibitors excl. heparin	ABCB1 CYP2C19 PTGS1 P2RY12	Toxicity Efficacy Efficacy;Toxicity Efficacy;Toxicity;Metabolism/PK Metabolism/PK	1A	May have increased platelet reactivity. May have increased risk of cardiovascular events. May have increased risk of Major Adverse Cardiac Events (MACE). May have decreased efficacy. May have increased risk of Stroke. May have increased risk of Mortality. May have increased non-responsiveness. May have decreased clopidogrel active metabolite level. May have decreased mean peak plasma concentration of H4. May have increased risk of Peripheral Vascular Diseases.
Efavirenz	Atripla Stocrin Sustiva Symfi	J05AG03	ANTIINFECTIVES FOR SYSTEMIC USE	Non-nucleoside reverse transcriptase inhibitors	CYP2B6 ABCB1	Toxicity Other Metabolism/PK Dosage	1A	May have increased plasma 4beta-OHC/Chol ratio. May have increased risk of Toxic liver disease. May have increased plasma level. May have increased plasma concentration. May have increased severity of Somnolence. May have increased likelihood of Sleep Disorders. May have increased PBMC-associated drug concentrations. May have increased risk of Central Nervous System Disorder. May have decreased likelihood of emerging viral drug resistance. May have increased likelihood of toxicity-related treatment failure.
Tacrolimus	Advagraf Astagraf Envarsus Modigraf Prograf Protopic	L04AD02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Calcineurin inhibitors	TCF7L2 CYP3A4 ABCB1 POR CYP3A5 CYP3A5;ZSCAN25 CAPN10	Toxicity Metabolism/PK Efficacy Dosage	1A	May have decreased trough concentrations. May have decreased concentration/dose ratio. May have increased risk of Diabetes Mellitus. May have decreased C/D ratio. May have increased blood pressure. May have increased risk of Drug Toxicity. May have increased risk of Nephrotoxicity. May increase the risk of Viral Diseases. May have increased formation of metabolites. May require increased time to reach stable dose.
Warfarin	Coumadin Jantoven	B01AA03	BLOOD AND BLOOD FORMING ORGANS	Vitamin K antagonists	VKORC1 CYP2C9 GGCX NQO1	Toxicity Efficacy Dosage	1A	May have increased risk of Hemorrhage. May require decreased warfarin dose. May have increased Blood Coagulation Disorders. May have increased mean INR during 3-month daily observation period in patients with mechanical heart valve replacement.

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								May have increased proportion of the first 6 months' treatment time spent within the target INR range (PTIR). May have increased mean AUC (area under the curve) prothrombin time. May have increased international normalised ratio variability (INR-var).
Amitriptyline	Elavil	N06AA09	NERVOUS SYSTEM	Non-selective monoamine reuptake inhibitors	CYP2C19	Metabolism/PK	1A	May have increased risk of treatment-emergent mania. May have increased amitriptyline/nortriptyline ratio corrected for dose and weight.
Capecitabine	Ecansya Xeloda	L01BC06	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Pyrimidine analogues	CDA ENOSF1 CES1P1 TYMS DPYD	Toxicity	1A	May have increased risk of Diarrhea. May have decreased likelihood of Asthenia. May have increased recurrence-free survival. May have increased likelihood of Nausea, Vomiting. May have decreased likelihood of Hand-foot syndrome. May have increased likelihood of Hyperbilirubinemia.
Citalopram	Celexa	N06AB04	NERVOUS SYSTEM	Selective serotonin reuptake inhibitors	CYP2C19 HTR2A	Metabolism/PK Toxicity	1A	May have increased intolerance. May have increased AUC of citalopram. May have increased median concentration/dose ratio and median parent drug/metabolite ratio. May have increased likelihood of Physiological sexual disorder.
Clomipramine	Anafranil	N06AA04	NERVOUS SYSTEM	Non-selective monoamine reuptake inhibitors	CYP2C19	Metabolism/PK	1A	May have increased risk of treatment-emergent mania. May have increased mean clomipramine concentrations corrected for dose and weight and higher metabolic ratio of clomipramine/desmethylclomipramine.
Escitalopram	Ciprallex Lexapro	N06AB10	NERVOUS SYSTEM	Selective serotonin reuptake inhibitors	HTR2A CYP2C19	Efficacy Metabolism/PK Toxicity	1A	May have increased serum concentration. May have increased antidepressant response. May have increased risk of adverse cognitive effects. May have increased risk of Physiological sexual disorder. May have increased likelihood of Discontinuation, Treatment modification. May have increased likelihood of suicidal ideation, adverse events, deliberate self-harm.
Fluorouracil	Actikerall Carac Efudex Fluoroplex Tolak Adrucil	L01BC02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Pyrimidine analogues	TYMS XRCC1 DPYD IGFBP3	Efficacy Other Toxicity	1A	May have better survival outcome. May have decreased Drug Toxicity. May have increased risk of Nausea. May have increased risk of Diarrhea.
Lansoprazole	Prevacid Prevpac	A02BC03	ALIMENTARY TRACT AND METABOLISM	Proton pump inhibitors	CYP2C19	Metabolism/PK Efficacy	1A	May have decreased intragastric pH. May have increased risk of Respiratory Tract Infections.

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Omeprazole	Konvomep Losec Omeclamox Omesec Previdolrx Prilosec	A02BC01	ALIMENTARY TRACT AND METABOLISM	Proton pump inhibitors	CYP2C19	Metabolism/PK Efficacy	1A	May have decreased Gastric pH. May have increased meal-stimulated plasma gastrin concentrations. May have decreased oral clearance (CL/F) of omeprazole racemate, 5-hydroxyomeprazole, R-omeprazole and S-omeprazole. May have increased area under the curve (AUC) of omeprazole racemate, omeprazole sulfone, R-omeprazole and S-omeprazole. May have increased metabolic ratio (defined as ratio of omeprazole to its metabolites) of 5-hydroxyomeprazole and R-5-hydroxyomeprazole.
Phenprocoumon	Phenprocoumon	B01AA04	BLOOD AND BLOOD FORMING ORGANS	Vitamin K antagonists	VKORC1 EPHX1	Dosage Efficacy Toxicity	1A	May have increased risk of adverse events. May require increased time to achieve a stable dose. May have increased likelihood of Hemorrhage.
Rosuvastatin	Crestor Ezallor Roszet	C10AA07	CARDIOVASCULAR SYSTEM	HMG CoA reductase inhibitors	ABCG2 FMO3	Efficacy Metabolism/PK	1A	May have increased plasma concentrations. May have increased likelihood of Elevated liver enzymes. May increase reduction in low-density lipoprotein cholesterol (LDL-C) levels in a gene-dose-dependent manner.
Sertraline	Zoloft	N06AB06	NERVOUS SYSTEM	Selective serotonin reuptake inhibitors	CYP2C19	Metabolism/PK	1A	May have increased risk of treatment-emergent mania. May have increased likelihood of Discontinuation, Treatment modification. May have increased mean serum concentrations of sertraline and N-desmethyl sertraline.
Voriconazole	Vfend	J02AC03	ANTIINFECTIVES FOR SYSTEMIC USE	Triazole derivatives	CYP2C19	Metabolism/PK	1A	May have decreased likelihood of Elevated liver enzymes, Hyperbilirubinemia, Toxic liver disease.
Alfentanil	Alfenta	N01AH02	NERVOUS SYSTEM	Opioid anesthetics	OPRM1	Dosage Toxicity Efficacy Metabolism/PK	3	May have increased risk of Pain. May have increased plasma concentration.
Allopurinol	Aloprim Zyloprim	M04AA01	MUSCULO-SKELETAL SYSTEM	Preparations inhibiting uric acid production	HLA-A	Toxicity	3	May have decreased risk of Severe Cutaneous Adverse Reactions.
Anastrozole	Arimidex	L02BG03	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Aromatase inhibitors	CYP19A1	Efficacy	3	May have increased recurrence-free survival.
Apixaban	Eliquis	B01AF02	BLOOD AND BLOOD FORMING ORGANS	Direct factor Xa inhibitors	ABCB1	Toxicity	3	May have decreased risk of Hemorrhage.
Aspirin	Aggrenox Alka seltzer	N02BA01	NERVOUS SYSTEM	Analgesic drugs	TBXA2R ADORA1	Toxicity Efficacy;Toxicity	3	May have increased risk of Death. May have decreased risk of aspirin-intolerant asthma.

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	Bayer Aspirin Duoplavin Durlaza Ecotrin Ecpirin Exaprin Excedrin Fasprin Orphengesic Vazalore Yosprala				PTGER2 NAT2 PTGIR TGFB1 HNMT IL1B HLA-DPB1 IL4 PEAR1			May have Peptic Ulcer Disease. May have increased risk of Hemorrhage. May have increased risk of intolerance. May have decreased platelet aggregation. May have increased risk of Myocardial Infarction. May have increased cardiovascular events in patients undergoing PCI.
Atazanavir	Evotaz Reyataz	J05AE08 J05AR23	ANTIINFECTIVES FOR SYSTEMIC USE	Protease inhibitors Antiviral drugs	CYP3A5	Metabolism/PK	3	May have increased oral clearance (CL) of atazanavir.
Atenolol	Tenoretic Tenormin	C07AB03	CARDIOVASCULAR SYSTEM	Beta blocking agents	PROX1 FTO	Toxicity	3	
Atorvastatin	Atorvaliq Caduet Lipitor Lypqozet	C10AA05	CARDIOVASCULAR SYSTEM	HMG CoA reductase inhibitors	KIF6 ABCG2 ABCB1 HMGCR SLCO1B1	Efficacy Toxicity Other	3	May have increased likelihood of myalgia. May have increased likelihood of Drug Toxicity, Muscular Diseases, Rhabdomyolysis, Toxic liver disease. May have LDL-C response. May have increased benefit. May have decreased plasma concentrations. May have decreased likelihood of statin-related myopathy. May have increased likelihood of Treatment interruption, Treatment modification, drug dosage alteration, Discontinuation.
Azathioprine	Azasan Imuran	L04AX01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Other immunosuppressants	ITPA DDR1;ITPA FTO	Efficacy Toxicity	3	May have increased likelihood of Leukopenia.
Bevacizumab	Avastin Mvasi	L01XC07	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Monoclonal antibodies	CXCR4	Efficacy	3	May have decreased Progression-free survival.
Buprenorphine	Belbuca Brixadi Buprenex Buprenorphine Butrans Sublocade Suboxone Subutex Zubsolv	N02AE01	NERVOUS SYSTEM	Drugs used in opioid dependence	OPRD1	Efficacy	3	May have decreased severity of Anxiety Disorders.

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Bupropion	Aplenzin Auvelity Budeprion Contrave Forfivo Wellbutrin Zyban	N06AX12	NERVOUS SYSTEM	Other antidepressants	CYP2B6 GALR1	Metabolism/PK Efficacy	2A	May have increased risk of Recurrence. May have decreased likelihood of smoking cessation. May have decreased hydroxybupropion concentrations and hydroxybupropion/bupropion ratio.
Caffeine	Anacin Exaprin Excedrin Fioricet Midol Migergot Norgesic Orbivan Orphengesic Peyona Trezix Trianal Triatec Vanatol Vanquish Vivarin	N06BC01	NERVOUS SYSTEM	Xanthine derivatives	ADORA2A	Toxicity	3	May increase caffeine toxicity.
Carvedilol	Coreg	C07AG02	CARDIOVASCULAR SYSTEM	Alpha and beta blocking agents	ADRB2	Efficacy	3	May have increased reduction in resting blood pressure. May have decreased likelihood of improved left ventricular ejection fraction.
Celecoxib	Celebrex Elyxyb Seglentsis	M01AH01	MUSCULO-SKELETAL SYSTEM	Coxibs	IL23R PTGES PTGER4	Toxicity	3	May have increased risk of Adenoma. May have increased risk of gastrointestinal toxicities. May have increased risk of cardiovascular toxicity and symptoms.
Cetuximab	Erbitux	L01XC06	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Monoclonal antibodies	EGFR	Efficacy	3	May have decreased Progression-free survival.
Cisplatin	Platinol	L01XA01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Platinum compounds	ERCC1 XPC TMEM43;XPC	Toxicity Efficacy	3	May have increased risk of Nephrotoxicity. May have increased severity of Ototoxicity. May have increased overall survival and progression-free survival.
Clozapine	Clozaril Fazaclo Versacloz	N05AH02	NERVOUS SYSTEM	Diazepines	ABCB1	Toxicity	3	May have increased Weight gain. May have increased Hypertension.

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Cocaine	Goprelto Numbrino	N01BC01	NERVOUS SYSTEM	Esters of benzoic acid	DRD2 OPRD1 OPRM1 CNR1	Toxicity Other	3	May have increased risk of Death.
Codeine	Ascomp Cheratussin Cheratussin Dac Codar Codeine Contin Covan	R05DA04	RESPIRATORY SYSTEM	Opium alkaloids and derivatives	ABCB1	Toxicity	3	May have increased likelihood of CNS depression in breast-feeding infants.
Cyclophosphamide	Procytox	L01AA01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Nitrogen mustard analogues	ABCC4 CYP2B6 CYP2C19	Toxicity	3	May have increased risk of Diarrhea. May have increased event-free survival. May have increased risk of Neutropenia. May have increased risk of oral mucositis. May have decreased likelihood of dose reduction. May have decreased risk of Graft-versus-host disease. May have increased risk of gastrointestinal toxicity. May have increased likelihood of Cystitis, Drug Toxicity. May have decreased risk of veno-occlusive disease of the liver.
Cyclosporine	Cequa Gengraf Neoral Restasis Sandimmune Verkazia Vevye	L04AD01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Immunosuppressants drugs	ABCB1	Metabolism/PK	3	May have decreased risk of delayed graft function. May have increased severity of Gingival Hyperplasia.
Cytarabine	Cytosar Vyxeos	L01BC01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Pyrimidine analogues	ABCB1 CDA	Efficacy Toxicity	3	May have increased Drug Toxicity. May have increased Overall survival.
Daunorubicin	Cerubidine Vyxeos	L01DB02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Anthracyclines and related substances	NOS3	Efficacy	3	May have increased Overall survival.
Desloratadine	Aerius Clarinx Neoclarityn	R06AX27	RESPIRATORY SYSTEM	Other antihistamines for systemic use	HRH1	Toxicity	3	May have increased severity of sedation.
Dexmedetomidine	Dexdor Igalmi Precedex	N05CM18	NERVOUS SYSTEM	Other hypnotics and sedatives	ADRA2A	Other	3	May have increased Bispectral Index and Ramsay Sedation Scores, indicating a longer period for falling asleep.

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Docetaxel	Docivyx Taxotere	L01CD02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Taxanes	ABCC2 SLCO1B3 VEGFA RXRA	Toxicity;Metabolism/PK Toxicity Other Efficacy	3	May have increased AUC. May have increased risk of Leukopenia. May have increased severity of Anemia. May have decreased progression-free survival.
Enalapril	Epaned Vaseretic Vasotec	C09AA02	CARDIOVASCULAR SYSTEM	ACE inhibitors	BDKRB2	Efficacy	3	May have increased risk of Cough.
Ethanol	Alcohol Ethanol	V03AZ01	VARIOUS	Nerve depressants	ANKK1;DRD2 DRD2 OPRM1 ADH1B PNPLA3	Toxicity Dosage Metabolism/PK	3	May have increased risk of Alcohol abuse. May have increased subjective feelings of intoxication, stimulation, sedation, and happiness.
Everolimus	Afinitor Votubia Zortress	L01EG02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Antineoplastic drugs	ABCB1	Toxicity	3	May have increased likelihood of Mucositis.
Fentanyl	Abstral Actiq Duragesic Effentora Fentora Instanyl Lazanda Sublimaze Subsys	N01AH01	NERVOUS SYSTEM	Anesthetic drugs	ADRB2 OPRM1	Other Toxicity Efficacy Dosage	2A	May have increased likelihood of Pain. May have increased likelihood of Nausea. May have increased likelihood of Somnolence. May have increased likelihood of Gastrointestinal toxicity. May have higher responsiveness to opioids and required less analgesic drugs in laboring women. May have increased severity of Hypotensive disorder.
Fluindione	Fluindione	B01AA12	BLOOD AND BLOOD FORMING ORGANS	Vitamin K antagonists	VKORC1	Efficacy	3	May require decreased average daily dose during the first period of stability.
Fluoxetine	Prozac Sarafem Symbyax	N06AB03	NERVOUS SYSTEM	Selective serotonin reuptake inhibitors	CRHR1	Efficacy	3	
Gefitinib	Iressa	L01EB01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Epidermal growth factor receptor (EGFR) tyrosine kinase inhibitors	IKBKE	Toxicity	3	May have increased risk of Exanthema.
Gemcitabine	Gemzar	L01BC05	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Pyrimidine analogues	SLC29A1 ABCG2 CDA RRM1 DCK	Toxicity Efficacy;Toxicity Other	3	May have increased likelihood of Nausea. May have increased Progression-free survival. May have increased severity of Drug Toxicity. May have increased risk of severe neutropenia. May have increased severity of Thrombocytopenia.

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								May have increased elimination clearance of dFdU and formation clearance of dFdCTP.
Hydrochlorothiazide	Accuretic Actelsar Hct Aldactazide Altace HCT Atacand Avalide Benicar Hct Diovan Hct Exforge Hct Hyzaar Ifirmacombi Karvezide Lopressor Hct Lotensin Hct Maxzide	C03AA03	CARDIOVASCULAR SYSTEM	Thiazides	ADD1 TCF7L2	Efficacy Toxicity	2B	May have increased risk of Diabetes Mellitus. May increase blood pressure reduction.
Hydrocodone	Dalmacol Hycet Hycodan Hydromet Hysingla Lorcet Lortab Norco Obredon Reprexain Tussicaps Tussionex Vicoprofen Xodol Zamiset Zohydro Zydone	R05DA03	RESPIRATORY SYSTEM	Opium alkaloids and derivatives	OPRM1	Toxicity	3	May have increased risk of Constipation. May have increased risk of side effects.
Imatinib	Gleevec Glivec	L01EA01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	BCR-ABL tyrosine kinase inhibitors	SLC19A1 SLC22A1 ABCB1 ABCG2 EGFR ABCB4 NQO1	Toxicity Efficacy;Toxicity;Metabolism/PK Efficacy Efficacy;Metabolism/PK Metabolism/PK	3	May have decreased severity of Drug Toxicity. May have increased severity of Exanthema. May have increased severity of Myelosuppression. May have increased risk of Drug Toxicity, Eye Diseases.

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Indinavir	Crixivan	J05AE02	ANTIINFECTIVES FOR SYSTEMIC USE	Protease inhibitors	UGT1A1	Toxicity	3	May have increased risk of Hyperbilirubinemia.
Irinotecan	Camptosar Onivyde	L01CE02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Topoisomerase 1 (TOP1) inhibitors	UGT1A1 ABCC1 TGFBI SLC01B1 UGT1A6 TGFBR2	Toxicity Efficacy Dosage	2A	May have increased risk of Diarrhea. May have increased risk of Neutropenia. May have increased risk of Drug Toxicity. May have increased severity of Leukopenia. May have increased Overall survival, Progression-free survival.
Isoniazid	Isonarif Isotamine Rifamate Rifater	J04AC01	ANTIINFECTIVES FOR SYSTEMIC USE	Hydrazides	MAFK	Toxicity	2A	May have increased likelihood of Toxic liver disease.
Lamotrigine	Lamictal	N03AX09	NERVOUS SYSTEM	Other antiepileptics	UGT2B7	Dosage Metabolism/PK	3	May have increased severity of drug-drug pharmacokinetic interaction.
Letrozole	Femara	L02BG04	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Aromatase inhibitors	ESR1	Toxicity	3	
Levodopa	Dhivy Duodopa Inbrija Parcopa Prolopa Rytary Sinemet Stalevo	N04BA01	NERVOUS SYSTEM	Dopa and dopa derivatives	DRD3	Toxicity	3	May have increased risk of Gastrointestinal toxicity.
Losartan	Cozaar Hyzaar	C09CA01	CARDIOVASCULAR SYSTEM	Angiotensin II receptor blockers (ARBs)	STK39 CAMK1D	Efficacy	3	May have decreased reduction in ambulatory blood pressure.
Melphalan	Alkeran Evomela Hepzato	L01AA03	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Nitrogen mustard analogues	SLC7A5	Toxicity	3	May have decreased risk of Gastrointestinal toxicity.
Mercaptopurine	Purixan Purinethol	L01BB02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Purine analogues	FTO	Toxicity	3	May have increased risk of Leukopenia.
Metformin	Actoplus Met Avandamet Fortamet	A10BA02	ALIMENTARY TRACT AND METABOLISM	Biguanides	SLC22A2 SLC47A2	Other Metabolism/PK Efficacy	3	May have increased renal and secretory clearance of metformin. May have increased likelihood of Drug Toxicity.

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	Glucophage Glucovance Glumetza Glycon Invokamet Janumet Jentadueto Kazano Kombiglyze Qternmet Riomet Segluromet Synjardy Trijardy Velmetia Xigduo Zituvimet							
Methadone	Diskets Dolophine Metadol Methadose	N07BC02	NERVOUS SYSTEM	Drugs used in opioid dependence	CYP3A4 CYP2B6 OPRM1	Toxicity Dosage Efficacy Metabolism/PK	2A	May have increased severity of Pain. May have increased likelihood of Postoperative Nausea and Vomiting. May have decreased likelihood of treatment. May have decreased likelihood of Recurrence. May have increased severity of side effects. May have increased severity of opioid withdrawal symptoms.
Methazolamide	Methazolamide	S01EC05	SENSORY ORGANS	Carbonic anhydrase inhibitors	HLA-C	Toxicity	2B	May have increased risk of Severe Cutaneous Adverse Reactions. May have increased risk of Toxic Epidermal Necrolysis, Stevens-Johnson Syndrome.
Methotrexate	Metoject Nordimet Otrexup Rasuvo Reditrex Trexall Xatmep	L01BA01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Other immunosuppressants	MTHFD1 SLC19A1 MTRR MTHFR DHFR ABCG2 ADORA2A ABCB1 ABCC2 ADORA2A;ADO RA2A-AS1 SLCO1B1 FPGS ABCC1 ITPA	Efficacy Toxicity Metabolism/PK Efficacy;Toxicity	2A	May have increased likelihood of Drug Toxicity. May have increased risk of Mucositis. May have increased risk of Nephrotoxicity. May have decreased survival. May have increased risk of Recurrence. May have increased risk of Discontinuation. May have increased likelihood of Treatment interruption. May have increased risk of prolonged high concentrations. May have increased speed of platelet recovery. May have increased plasma level.

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Methylphenidate	Aptensio Biphentin Concerta Cotempla Daytrana Foquest Jornay Metadate Methylin Quillichew Quillivant Relexxii Ritalin	N06BA04	NERVOUS SYSTEM	Centrally acting sympathomimetics	DRD3	Efficacy	3	
Morphine	Arymo Avinza Duramorph Embeda Infumorph Kadian MSIR Mitigo Contin Statex	N02AA01	NERVOUS SYSTEM	Natural opium alkaloids	FAAH OPRM1	Toxicity Dosage Efficacy	3	May have decreased likelihood of treatment. May have increased risk of Postoperative Nausea and Vomiting.
Naloxone	Kloxxado Narcan Suboxone Targin Targiniq Zimhi Zubsolv	V03AB15	ALIMENTARY TRACT AND METABOLISM	Peripheral opioid receptor antagonists	OPRM1	Efficacy	3	May have increased cortisol response.
Naltrexone	Contrave Embeda Vivitrol	N07BB04	NERVOUS SYSTEM	Drugs used in alcohol dependence	OPRD1	Efficacy	3	May have increased naltrexone-induced blunting of alcohol stimulation and alcohol craving.
Nevirapine	Viramune	J05AG01	ANTIINFECTIVES FOR SYSTEMIC USE	Non-nucleoside reverse transcriptase inhibitors	ABCB1 CYP2B6 CCHCR1 POLR3G	Toxicity Metabolism/PK	2A	May have increased risk of Exanthema. May have increased plasma level. May have increased plasma concentration. May have increased risk of Drug Toxicity. May have decreased risk of Toxic liver disease. May have increased risk of Nevirapine-induced rash.

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Nicotine	Habitrol Nicoderm C-Q Nicorelief Nicorette Nicotrol	N07BA01	NERVOUS SYSTEM	Drugs used in nicotine dependence	CHRNA4 OPRM1 DRD1 DDC ANKK1;DRD2 CHRNA4 DRD2 CHRNA3 CHRNA3;CHRNA5 ANKK1 GRIN3A	Other Efficacy Toxicity Efficacy;Toxicity	3	May have increased likelihood of smoking cessation. May have increased severity of Tobacco Use Disorder. May have decreased risk of Substance-Related Disorders. May have increased likelihood of nicotine dependence.
Nortriptyline	Aventyl Pamelor	N06AA10	NERVOUS SYSTEM	Non-selective monoamine reuptake inhibitors	GNB3	Efficacy	3	May have Weight gain. May have improvements in neurovegetative symptoms. May have decreased likelihood of Sleep Initiation and Maintenance Disorders.
Olanzapine	Lybalvi Olazax Symbyax Zalasta Zypadhera Zyprexa	N05AH03	NERVOUS SYSTEM	Diazepines	HTR2C ABCB1 DRD2 CYP3A5 AHR	Toxicity Other Metabolism/PK	3	May have increased prolactin. May have increased Weight gain. May have increased area under the curve. May have increased likelihood of Somnolence. May have 29% lower mean 4'-desmethyl OLA/olanzapine (OLA) ratio. May have positive relationship between olanzapine plasma levels and positive symptom reduction compared to subjects with the GG genotype.
Oseltamivir	Ebilfumin Tamiflu	J05AH02	ANTIINFECTIVES FOR SYSTEMIC USE	Neuraminidase inhibitors	ABCB1	Toxicity	3	May have increased risk of Depression, Gastritis, Hypersensitivity.
Oxaliplatin	Oxaliplatin	L01XA03	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Platinum compounds	ABCB1	Efficacy	3	May have increased recurrence-free survival. May have increased likelihood of Peripheral Nervous System Diseases.
Oxcarbazepine	Oxtellar Trileptal	N03AF02	NERVOUS SYSTEM	Carboxamide derivatives	HLA-B	Toxicity	3	May have increased risk of Maculopapular Exanthema.
Oxycodone	Endocet Nalocet Oxaydo Oxycontin Oxyneo Percocet Prolate Rivacocet	N02AA05	NERVOUS SYSTEM	Natural opium alkaloids	OPRM1 ABCB1	Dosage Efficacy	3	May have increased likelihood of Pain. May have increased likelihood of adverse events.

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	Roxicet Roxicodone Roxybond Targin Targiniq Xolox Xtampza							
Paclitaxel	Abraxane Taxol	L01CD01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Taxanes	EPHA5 CYP2C8 CYP3A5	Toxicity	3	May have increased risk of Anemia. May have decreased risk of Neurotoxicity Syndromes. May have increased risk of Leukopenia, Neutropenia. May have increased risk of Peripheral Nervous System Diseases.
Peginterferon Alfa-2B	Pegintron Sylatron	L03AB10	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Interferons	ITPA	Toxicity	2B	May have decreased anaemia. May have increased sustained virological response (SVR).
Pemetrexed	Alimta Ciambra Pemfexy Pemrydi Rtu	L01BA04	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Folic acid analogues	DHFR MTHFR	Toxicity Efficacy	3	May have decreased Overall survival. May have increased risk of Drug Toxicity.
Phenytoin	Dilantin Phenytek	N03AB02	NERVOUS SYSTEM	Hydantoin derivatives	ABCB1	Dosage	3	May have increased plasma drug levels. May have increased likelihood of drug resistance.
Prasugrel	Effient	B01AC22	BLOOD AND BLOOD FORMING ORGANS	Platelet aggregation inhibitors excl. heparin	CYP2C19	Efficacy	3	May have increased risk of high on-treatment platelet reactivity.
Pravastatin	Pravachol	C10AA03	CARDIOVASCULAR SYSTEM	HMG CoA reductase inhibitors	KIF6 HMGCR CETP	Efficacy	2B	May have increased risk of Death. May have decreased risk of Coronary Disease. May have decreased risk of Myocardial Infarction. May have decreased reduction in total cholesterol.
Repaglinide	Enyglid Gluconorm Prandin	A10BX02	ALIMENTARY TRACT AND METABOLISM	Other blood glucose lowering drugs	SLC01B1	Other	3	May have decreased repaglinide plasma concentration (AUC) and Cmax.
Ribavirin	Ibavir Rebetol Virazole	J05AP01	ANTIINFECTIVES FOR SYSTEMIC USE	Antivirals for treatment of HCV infections	ITPA	Toxicity	2B	May have decreased anaemia. May have increased sustained virological response (SVR).
Risperidone	Perseris Risperdal Rykindo Uzedy	N05AX08	NERVOUS SYSTEM	Other antipsychotics	ABCB1 UGT1A1 ANKK1;DRD2 DRD3 HTR2A	Toxicity Efficacy	3	May have increased severity of Weight gain. May have increased risk of Hyperprolactinemia. May have increased risk of extrapyramidal symptoms.

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Name	Trade Name	ATC	ATC Group	ATC Subgroup	Gene	Impact	Evidence	Clinical Effect
Ritonavir	Kaletra Norvir Paxlovid Viekira	J05AE03	ANTIINFECTIVES FOR SYSTEMIC USE	Protease inhibitors	APOC3	Toxicity	3	May have increased triglyceride levels.
Rosiglitazone	Avandamet Avandia	A10BG02	ALIMENTARY TRACT AND METABOLISM	Thiazolidinediones	PLIN1	Toxicity	3	May have decreased Weight gain.
Sevoflurane	Sevorane Sojourn Ultane	N01AB08	NERVOUS SYSTEM	Halogenated hydrocarbons	FASTKD3;MTRR CYP2E1 CYP2E1;DUX1 GABRA2	Efficacy Other	3	May have decreased mean arterial pressure. May have increased vol% end-tidal sevoflurane concentration.
Simvastatin	Cholib FloLipid Simcor Vytorin Zocor	C10AA01	CARDIOVASCULAR SYSTEM	HMG CoA reductase inhibitors	ABCG2 ABCB1 CYBA	Metabolism/PK Toxicity Other	3	May have decreased risk of Myalgia. May have increased reduction in DNA damage. May have increased likelihood of Toxic liver disease.
Sorafenib	Nexavar	L01EX02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Other protein kinase inhibitors	VEGFA NOS3 ABCB1 PRKCE SLCO1B1 KDR	Toxicity Efficacy	3	May have increased risk of Hypertension. May have increased risk of Hand-foot syndrome. May have decreased Overall survival. May have decreased likelihood of Diarrhea. May have increased likelihood of Hyperbilirubinemia.
Spironolactone	Aldactazide Aldactone Carospir	C03DA01	CARDIOVASCULAR SYSTEM	Aldosterone antagonists	ACE	Efficacy	3	May improve left ventricular ejection fraction and end-systolic and end-diastolic volumes.
Sufentanil	Dsuvia Sufenta	N01AH03	NERVOUS SYSTEM	Opioid anesthetics	OPRM1	Dosage	3	May have decreased severity of Pain. May have increased likelihood of adverse events.
Sunitinib	Sutent	L01EX01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Other protein kinase inhibitors	SLCO1B3 ABCB1 NR112 CXCL8 ABCG2 FLT3 CYP3A5	Efficacy Toxicity Dosage;Toxicity	3	May have decreased Overall survival. May have decreased risk of Diarrhea. May have increased risk of Hypertension. May have increased severity of Thrombocytopenia. May have increased risk of Leukopenia, Neutropenia.
Tamoxifen	Soltamox	L02BA01	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Anti-estrogens	ESR1 ABCB1	Toxicity Efficacy	3	May have increased risk of disease recurrence.
Ticagrelor	Brilinta Brilique	B01AC24	BLOOD AND BLOOD FORMING ORGANS	Platelet aggregation	PEAR1	Efficacy	3	May have increased risk of Hemorrhage.

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Name	Trade Name	ATC	ATC Group	ATC Subgroup	Gene	Impact	Evidence	Clinical Effect
				inhibitors excl. heparin				
Tramadol	Conzip Durela Qdolo Ralivia Ryzolt Seglensis Tridural Ultracet Ultram Zytram	N02AX02	NERVOUS SYSTEM	Other opioids	ABCB1 OPRM1	Efficacy Metabolism/PK Toxicity	3	May have increased likelihood of Nausea. May have increased likelihood of Constipation. May have increased likelihood of Gastrointestinal toxicity.
Valproic Acid	Depakene Depakote Epival	N03AG01	NERVOUS SYSTEM	Fatty acid derivatives	UGT1A10;UGT1A6;UGT1A7;UGT1A8;UGT1A9 ANKK1 UGT1A6	Dosage Toxicity	3	May have increased Weight gain.
Verapamil	Calan Isoptin Tarka Verelan	C08DA01	CARDIOVASCULAR SYSTEM	Phenylalkylamine derivatives	NR1H3	Toxicity	3	May have decreased risk of myocardial infarction and stroke.
Verapamil	Calan Isoptin Tarka Verelan	C08DA01	CARDIOVASCULAR SYSTEM	Phenylalkylamine derivatives	NR1H3	Toxicity	3	May have decreased risk of myocardial infarction and stroke.
Vincristine	Marqibo Vincasar	L01CA02	ANTINEOPLASTIC AND IMMUNOMODULATING AGENTS	Vinca alkaloids and analogues	CAPG CEP72	Toxicity	3	May have decreased risk of Neurotoxicity Syndromes. May have increased risk of Peripheral Nervous System Diseases.

5 GENETIC AND MOLECULAR RESULTS

5.1 Detected Diplotypes with Altered Functions

The results of the genetic analysis are presented using the patient's "Detected Diplotypes." A diplotype is the specific combination of two alleles (versions of a gene) inherited by the patient.

5.2 Understanding Alleles and Star Nomenclature

To standardise reporting, most pharmacogenes use the "star (*)" nomenclature system, which was originally developed at the Karolinska Institutet and is now maintained by PharmVar.

- **Allele:** Usually refers to the "normal" or "wild-type" version of the gene, which produces a protein with normal activity levels.
- **Variant Alleles:** These represent genetic changes that may result in increased activity, decreased activity, or complete loss of function.

5.3 Predicting the Phenotype

The combination of two alleles (the diplotype) determines the patient's overall phenotype—their observable metabolic status.

Predicted Phenotype	Genetic Basis	Expected Drug Response
Ultrarapid Metaboliser	Two alleles with increased activity or extra copies of the gene.	Very high clearance; high risk of treatment failure for many drugs.
Normal Metaboliser	Two "wild-type" or normal alleles.	Standard response and clearance; follows normal dosing guidelines.
Intermediate Metaboliser	One normal allele and one decreased-function allele.	Moderately reduced clearance; may require slight dose reductions.
Poor Metaboliser	Two alleles that have little or no function.	Very low clearance; high risk of toxicity at standard doses.

In some complex cases, such as the gene, an "Aggregated Phenotype" is reported. This accounts for the fact that a patient may have more than two functional copies of a gene (due to gene duplication) or may have a combination of several different alleles that work together to determine the overall metabolic rate.

5.4 Detected Gene's Haplotypes for All Tiers in the Association for Molecular Pathology Extended List (AMP)

Gene	Genotype	Effect
CYP2A6	*1/*5	Increased Function Rapid Metaboliser
CYP2B6	*1/*6	Decreased Function Intermediate Metaboliser
CYP2C19	*1/*2	Decreased Function Intermediate Metaboliser
CYP2D6	*1/*2	Normal Function Normal Metaboliser
CYP3A5	*1/*3	Decreased Function Intermediate Metaboliser(oxycodone)
HLA-A	*02:01/*02:01	Increased Risk of Adverse Drug Reactions
HLA-B	*1/*38:02	Increased Risk of Adverse Drug Reactions
HLA-C	*1/*01:02	Increased Risk of Adverse Drug Reactions
HLA-DRB1	*15:01/*15:01 *16:02/*16:02	Increased Risk of Adverse Drug Reactions
NAT2	*1/*7A *1/*7B *1/*13A *12A/*12A	Indeterminate Function Indeterminate Metaboliser***
UGT1A1	*1/*6	Decreased Function Intermediate Metaboliser
UGT1A3	*1/*2	Increased Function Rapid Metaboliser

Effect column:

Where more than 1 diplotype is present for a gene, aggregated phenotypes of all alleles are shown.

***Aggregated phenotype of all alleles: 12A: Rapid Metaboliser, 7A and 7B are Poor Metabolisers.

6 DISCLAIMERS

Technical Methodology and Validation Standards

The accuracy of a pharmacogenetic report depends on the quality of the laboratory analysis. This test uses advanced molecular biology techniques to identify genetic variants.

1 Laboratory Technologies

Most clinical pharmacogenetic tests are performed using:

- **Array-based Assays:** These use a silicon chip to simultaneously detect hundreds of known genetic variants with very high precision (sensitivity and specificity >99%).
- **Next-Generation Sequencing (NGS):** This technology allows for the "reading" of large sections of the genome, including both common and rare variants. NGS is particularly useful for identifying rare variants that may not be present on standard array chips.

2 Compliance with Technical Standards

The laboratory analysis follows the guidelines set by the American College of Medical Genetics and Genomics (ACMG) and the Association for Clinical Genetic Science (ACGS) in the UK. These standards ensure that:

- **Accuracy:** The test correctly identifies the genetic variants present in the patient's sample.
- **Nomenclature:** Variants are named using meaningful, unambiguous, and consistent terms (e.g., star alleles).
- **Authoritative Interpretation:** The clinical implications are derived from peer-reviewed literature and consensus guidelines.

Clinical Caveats and Disclaimers: Understanding the Limits of Genomics

While pharmacogenetics is a powerful tool, it is not a perfect predictor of drug response. It provides one piece of a complex clinical puzzle.

3 The Role of Non-Genetic Factors

The "expressed phenotype"—how a patient actually responds to a drug—is influenced by many factors beyond genetics.

- **Phenoconversion:** This occurs when external factors change a patient's metabolic status. For instance, a patient who is genetically a "Normal Metaboliser" can become a functional "Poor Metaboliser" if they take a medication like paroxetine or fluoxetine, which blocks the enzyme.
- **Drug-Drug Interactions:** Most patients take multiple medications, which can interfere with one another through shared metabolic pathways.
- **Lifestyle and Comorbidities:** Factors such as smoking, alcohol consumption, liver or kidney disease, age, and pregnancy can all significantly alter how the body handles medication.

4 Genetic Testing Limitations

- **Undetected Variants:** Genetic tests only look for the specific alleles they are programmed to find. If a patient has an extremely rare or novel mutation that is not on the test panel, the test will return a "normal" result even if the patient's metabolism is actually abnormal.
- **Complex Gene Structures:** Some genes, like those, are prone to structural variations, including gene deletions, duplications, or the formation of "hybrid" genes. In rare cases, these complex structures can make it difficult to determine the exact phenotype.
- **Tissue Differences:** For patients who have received a liver or bone marrow transplant, a DNA test (which usually uses blood or saliva) may not reflect the genetics of the organs responsible for drug metabolism.

5 Important Legal and Medical Information

- **Not a Replacement for Clinical Care:** Pharmacogenetic testing does not replace the need for therapeutic drug monitoring or careful clinical observation by a doctor.
- **No Changes without Consultation:** Patients must never change their medication or dosage based on this report without first consulting their prescribing physician or pharmacist.
- **General Health Information:** The content of this report is provided as educational health information and is not a substitute for professional medical advice or diagnosis.
- **Regulatory Evaluation:** The reporting software used for these associations may not have been evaluated by all national regulatory agencies (such as the FDA), and the report is not intended to cure or prevent any disease.
- **Data Privacy:** All genetic data is kept confidential and protected by laws such as the Genetic Anti-Discrimination Act, which prevents insurers or employers from discriminating against individuals based on their genetic information.